Face recognition camera

API interface documentation

collection interface

collection interface			
order number	function	URI	Request method
1	Command Query	/VIID/System/Keepalive	POST
2	Get portrait set	/VIID/Extend/ExtendFaceList	GET
3	Report Command Execution Status	/VIID/Extend/ExtendConfirm	
4	Report Device Portrait Collection	/VIID/Extend/ExtendReportFaceList	
5	Report Facial Recognition Data	/VIID/Extend/ExtendFaceRecognition	

1. Command Query

1. Interface Overview

URI	/VIID/System/Keepalive		
method	**Query string**	**message body**	**Return result**
POST	not have	< Keepalive >	<ResponseStatus>
remarks			

2. Message Body Feature Parameters

Reference standard /VIID/System/Keepalive interface

3. Return the result

Reference standard /VIID/System/Keepalive interface

Add the ExtendCmd field to the response body. Refer to A.7 for field definition.

4. instance

URI	/VIID/System/Keepalive
Request method	POST
request header	See request parameters
request body	{"KeepaliveObject":{"DeviceID":"33010299011190000253"}}

responder	{
	"ResponseStatusObject": {
	"RequestURL": "/VIID/System/Keepalive",
	"StatusCode": 0,
	"StatusString": "success",
	"ID": "33010299011190000253",
	"ExtendCmd":10001
	}
	}

2. Get the portrait collection (face information)

1. Interface Overview

URI	/VIID/Extend/ExtendFaceList		
method	**parameter**	**message body**	**Return result**
GET	? DeviceID=34020000001320000086&Size=5		
remarks	Size=5 means 5 face images are retrieved at once		

2. Message Body Feature Parameters

not have

3. Return the result

Refer to A.8 for the structure and A.9 for the field definitions.

Note: If Size=5 and the PersonListObject->Person data is less than 5, the face data will complete the request.

If Size=5 and the PersonListObject->Person data count is 5, the device will also call ExtendFaceStorage to request data.

PersonListObject->Person can only be <=Size

4. instance

URI	/VIID/Extend/ExtendFaceList?DeviceID=34020000001320000086&Size=5
Request method	GET
request header	See request parameters
request body	

responder	{
	"PersonListObject": {
	"Person": [
	{
	"TaskID": "1441307904689930241",
	"PersonID": "11010000001190000001022017121910051599983015647 9",
	"Name": "4V8FsvrVYGWIXl9cVUvk/g==",
	"GroupID": "03742882aefb11eb9cea002ec777281b",
	"Type": 0,
	"SubImageList":{
	"SubImageInfoObject":[
	{
	"Data": "base64-encoded jpg face image"
	"FileFormat":"Jpeg",
	"ImageID":"33020300001190000001022017122111111100011",
	"ShotTime":"20211110104853",
	"StoragePath":null,
	"Type":"11"
	}
	]
	}
	} ,
	{
	"ID": "1441307904689930242",
	"PersonID": "11010000001190000001022017121910051599983015647 8",
	"Name": "4V8FsvrVYGWIXl9cVUvk/g==",
	"GroupID": "03742882aefb11eb9cea002ec777281b",
	"Type": 0,
	"SubImageList":{
	"SubImageInfoObject":[

```
                    {
                        "Data": "base64-encoded jpg face image"
                        "FileFormat":"Jpeg",
                        "ImageID":"33020300001190000001022017122111111100011",
                        "ShotTime":"20211110104853",
                        "StoragePath":null,
                        "Type":"11"
                    }
                ]
            }
        }
    ]
    }
}
```


3-Report the command execution status

1. Interface Overview

URI	/VIID/Extend/ExtendConfirm		
method	**parameter**	**message body**	**Return result**
POST		< ConfirmObject >	<ResponseStatusList>
remarks	A report containing the results of one or more instruction executions		

2. Message Body Feature Parameters

Refer to A.10 for the message body structure and A.11 for field definitions.

Parameter name	**type**	**Mandatory items**	**Windows default**	**description**
TaskID	String	Y		Unique identifier for portrait set objects, see A.9 Portrait Set Object Definition

Structure reference:

```
{
"ConfirmListObject": {
"ConfirmObject": [
        {
          "TaskID" :"1453635655379718145",
          "StatusCode": 0,
          "StatusString": "success"
          },
          {
          "TaskID" :"1453635655379718146",
          "StatusCode": 0,
          "StatusString": "success"
          },
    ]
  }
}
```


3. Return the result

Refer to A.21 for the structure and A.22 for the field definitions.

4. instance

URI	/VIID/Extend/ExtendConfirm
Request method	POST
request header	See request parameters

request body	{
	"ConfirmListObject": {
	"ConfirmObject": [
	{
	"TaskID" :"1453635655379718145",
	"StatusCode": 0,
	"StatusString": "success"
	},
	{
	"TaskID" :"1453635655379718146",
	"StatusCode": 0,
	"StatusString": "success"
	}
	]

	} }

responder	{
	"ResponseStatusListObject": {
	"ResponseStatusObject": [
	{
	"RequestURL": "/VIID/Extend/ExtendConfirm",
	"StatusCode": "0",
	StatusString: "Normal"
	"Id": "33010101001210000001022017101808060802206036985 6",
	"LocalTime": "20171222104652"
	}
	]
	}
	}

4-Report Device Portrait Collection

1. Interface Overview

URI	/VIID/Extend/ExtendReportFaceList		
method	**parameter**	**message body**	**Return result**
POST		< FaceReportListObject >	<ResponseStatusList>
remarks			

2. Message Body Feature Parameters

Refer to A.12 for the message body structure and A.13 for field definitions.

3. Return the result

Refer to A.21 for the structure and A.22 for the field definitions.

4. instance

URI	/VIID/Extend/ExtendReportFaceList
Request method	POST

request header	See request parameters

request body	{
	" FaceReportListObject ": {
	"FaceReportObject": [
	{
	"PersonID": "11010000099830156479",
	"Name": "4V8FsvrVYGWIXl9cVUvk/g==",
	"GroupID": "03742882aefb11eb9cea002ec777281b",
	},
	{
	"PersonID": "11010000099830156478",
	"Name": "4V781VYGWIXl9cVUvk/g==",
	"GroupID": "03742882aefb11eb9cea002ec777281c",
	}
	]
	}
	}

responder	{
	"ResponseStatusListObject": {
	"ResponseStatusObject": [
	{
	"RequestURL": "/VIID/Extend/ExtendReportFaceList",
	"StatusCode": "0",
	StatusString: "Normal"
	"Id": "33010101001210000001022017101808060802206036985 6",
	"LocalTime": "20171222104652"
	}
	]
	}
	}

5. Submit face recognition data

1. Interface Overview

URI	/VIID/Extend/ExtendFaceRecognition		
method	**parameter**	**message body**	**Return result**

POST		< FaceRecognitionResultList >	<ResponseStatusList>
remarks	Image information is the capture result (i.e., consistent with the image information defined by the standard protocol capture interface)		

2. Message Body Feature Parameters

Refer to A.10 for the message body structure and A.11 for field definitions.

3. Return the result

Refer to A.21 for the structure and A.22 for the field definitions.

4. instance

URI	/VIID/Extend/ExtendFaceRrecognition
Request method	POST
request header	See request parameters

request body	{
	" PersonRecognitionResultListObject ": {
	"PersonRecognitionObject": [
	{
	"PersonID": "11010000099830156479",
	"Name": "4V8FsvrVYGWIXl9cVUvk/g==",
	"GroupID": "03742882aefb11eb9cea002ec777281b",
	"DeviceID":"55000111221190000007",
	"RecognitionTime":"20210910175723",
	"Similarity":"0.99"
	//Standard SubImageList
	SubImageList...
	}
	]
	}
	}

responder	{
	"ResponseStatusListObject": {
	"ResponseStatusObject": [
	{
	"RequestURL": "/VIID/Extend/ExtendFaceRrecognition",
	"StatusCode": "0",
	StatusString: "Normal"
	"Id": "33010101001210000001022017101808060802206036985 6",
	"LocalTime": "20171222104652"
	}
	]
	}
	}

Appendix A

View Library Object Feature Properties

A.7 ExtendCmd Extended Field Definition

The characteristics of the extended registration are shown in Table A.8.1.

Table A.8.1 Extended Registration Object Characteristics

order Number	name	ID	type XML/JSON	length	required / selectable	remarks
1	extened instruction	ExtendCmd	int		R/O	10001: Update the face database (including additions, modifications, and partial deletions)

A.8 Portrait collection object

```
<complexType name="Person">
    <sequence>
        <element name="TaskID" type="String" use="required">
        <element name="PersonID" type="String" use="required">
        <element name="Name" type="String" AES="required">
        <element name="GroupID" type="String" use="required">
        <element name="Type" type="Integer" use="required">
        <element name="SubImageList" type="SubImageInfoList" use="required">
    </sequence>
</complexType>
```

The Person set is the collection of Person objects.

```
<complexType name="PersonList">
    <sequence>
        <element name="Person" type="Person" minOccurs="1" />
    </sequence>
</complexType>
```

A.9 Portrait Collection Object Definition

The characteristic attributes of the portrait collection objects are shown in Table A.9.1.

Table A.9.1 Portrait Set Object Feature Attributes

Table A.9.1 Portrait Set Object Feature Attributes

order	name	ID	type XML/JSON	length	required /	remarks
1	task unique	TaskID	String	32	R	task unique identifier
2	Person	PersonID	String	200	R	Person identification
3	surname and	Name	String	64	O	surname and personal name
4	Member portrait	GroupID	String	200	R	Portrait ID
5	data type	Type	Integer		R	Data type (0: Add; 1: Modify; 2: Delete), default is 0
6	Image List	SubImageList	SubImageInfoList		R	It can contain 0 or more sub-image objects. Refer to the SubImageInfoList extension type in the protocol.

A.10 Reporting the execution status of the command object

```
<complexType name="Confirm">
    <sequence>
        <element name="TaskId" type="String" use="required"/>
        <element name="StatusCode" type=" Integer " use="required"/>
        <element name="StatusString" type="String"/>
    </sequence>
</complexType>

// Check if the ID object set is the ConfirmId object set
<complexType name="ConfirmList">
    <sequence>
        <element name="ConfirmObject" type="Confirm" minOccurs="1">
    </sequence>
</complexType>
```

A.11 Reporting the execution status of command objects

The characteristic attributes of the reporting command execution status object are shown in Table A.11.1.

Table A.11.1 Characteristics of Instruction Execution Status Attributes

Table A.11.1 Characteristics of Instruction Execution Status Attributes

order	name	ID	type JSON	length	Required/	remarks
1	assignment Id	TaskId	String	32	R	task unique identifier
2	Execution	StatusCode	int		R	0: Success Other failures

3	remarks	StatusString	String	500	R/O	Success: Success. Failure: Failure reason.

A.12 Reporting of Portrait Collection Objects

```
<complexType name="FaceReport">
    <sequence>
        <element name="PersonID" type="String" use="required">
        <element name="Name" type="String" AES="required">
        <element name="GroupID" type="String" use="required">
    </sequence>
</complexType>
```

A.13 Definition of Portrait Collection Objects for Reporting

The characteristic attributes of the subjects in the report are shown in Table A.13.1.

Table A.13.1 Reported Portrait Collection Object Attributes

Table A.13.1 Reported Portrait Collection Object Attributes

order	name	ID	type XML/JSON	length	required /	remarks
1	Person	PersonID	String	200	R	Person identification
2	surname and	Name	String	64	O	surname and personal name
3	Member portrait	GroupID	String	200	R	Portrait ID

A.21 Response Status Object

The characteristic attributes of the response status object are shown in Table A.21.1.

Table A.21.1 Response Status Object Feature Attributes

NO.r	name	ID	type XML/JSON	length	Required/ Optional	remarks
1	resource	RequestURL	string	256	R	URL for the corresponding operation

2	condition code	StatusCode	ConfirmStatusType		R	O-OK， normal 1-OtherError, other unknown error 2-Device Busy, device is busy 3-Device Error 4-Invalid Operation 5-Invalid XML Format, the XML format is invalid 6-Invalid XML Content, the XML content is invalid 7-Invalid JSON format. The JSON format is
3	state	StatusString	string	256	R	
4	resource ID	Id	string	64	R/O	The POST method returns an ID when creating a resource. After successful creation A new ID must be returned. If creation fails, this ID will not be available.
5	Date and Time	LocalTime	dateTime		O	Current time, used for time synchronization

A.22 Response Status Object

```
//Response status object
<complexType name="ResponseStatus">
    <sequence>
        <element name="RequestURL" type="string " use="required"/>
        <element name="StatusCode" type=" int" use="required"/>
        <element name="StatusString" type="string" />
        <element name="Id" type="string" minOccurs="0" maxOccurs="1"/>
        <element name="LocalTime" type="dateTime" minOccurs="0"
        maxOccurs="1"/>
    </sequence>
</ complexType>

//Response status object list
<complexType name="ResponseStatusList">
    <sequence>
        <element name="ResponseStatusObject" type="ResponseStatus" minOccurs="1"
        />
    </sequence>
</complexType>
```